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import plotly.graph_objects as go
monthly_sales_trend = df.set_index("Order
Date").resample("M")["Sales"].sum().reset_index()
fig = go.Figure(
    go.Scatter(
        x=monthly_sales_trend["Order Date"],
        y=monthly_sales_trend["Sales"],
        mode='lines',
        name='Monthly Sales',
        hovertemplate='<b>Date:</b> %{x|%Y-%m}<br><b>Sales:</b>
₹%{y:,.2f}'
    )
)

fig.update_layout(
    title="Monthly Sales Trend Over Time",
    xaxis_title="Order Month",
    yaxis_title="Total Sales"
)

fig.show()
category_segment = df.groupby(
    ["Category", "Segment"],
    as_index=False
)["Sales"].sum()

fig = go.Figure()

for segment_name in category_segment["Segment"].unique():
    segment_data = category_segment[category_segment["Segment"] ==
segment_name]
    fig.add_trace(
        go.Bar(
            x=segment_data["Category"],
            y=segment_data["Sales"],
            name=segment_name,
            hovertemplate='<b>Category:</b> %{x}<br><b>Segment:</b> ' +
segment_name + '<br><b>Sales:</b> ₹%{y:,.2f}'

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    )
)

fig.update_layout(
    barmode='group',
    title="Sales by Category and Customer Segment",
    xaxis_title="Category",
    yaxis_title="Sales"
)

fig.show()

fig = go.Figure(
    go.Scatter(
        x=df["Sales"],
        y=df["Profit"],
        mode='markers',
        marker=dict(
            size=df["Quantity"],
            sizemode='area',
            sizeref=2.*max(df["Quantity"])/(40.**2),
            sizemin=4,
            color=df["Category"].astype('category').cat.codes,
            colorscale='Viridis',
            showscale=True
        ),
        text=df["Product Name"],
        customdata=df[['Category']].values, # Pass Category as custom data
        hovertemplate=
            '<b>Product:</b> %{text}<br>' + # Product Name
            '<b>Sales:</b> ₹%{x:,.2f}<br>' + # Sales
            '<b>Profit:</b> ₹%{y:,.2f}<br>' + # Profit
            '<b>Quantity:</b> %{marker.size}<br>' + # Quantity
            '<b>Category:</b> %{customdata[0]}'
    )
)

fig.update_layout(
    title="Sales vs Profit (Plotly Go)",
    xaxis_title="Sales",

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        yaxis_title="Profit"
    )
fig.show()

fig = go.Figure()
for category in df["Category"].unique():
    category_data = df[df["Category"] == category]
    fig.add_trace(
        go.Histogram(
            x=category_data["Sales"],
            name=category,
            opacity=0.75,
            hovertemplate=
                '<b>Category:</b> ' + category + '<br>' + # Category
                '<b>Sales Range:</b> %{x}<br>' + # Sales Range
                '<b>Count:</b> %{y}' # Count
        )
    )
fig.update_layout(
    barmode='overlay',
    title="Distribution of Sales by Category (Histogram - Plotly Go)",
    xaxis_title="Sales",
    yaxis_title="Count"
)
fig.show()

fig = go.Figure()

for category in df["Category"].unique():
    category_data = df[df["Category"] == category]
    fig.add_trace(
        go.Box(
            y=category_data["Profit"],
            name=category,
            boxpoints='outliers',
            customdata=category_data[["Segment"]].values,
            hovertemplate=
                '<b>Category:</b> ' + category + '<br>' + # Category

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        '<b>Profit:</b> ₹%{y:,.2f}<br>' + # Profit
        '<b>Segment:</b> %{customdata[0]}'
    )
)
fig.update_layout(
    title="Profit Distribution Across Categories (Box Plot - Plotly Go)",
    xaxis_title="Category",
    yaxis_title="Profit"
)
fig.show()

fig = go.Figure()
for category in df["Category"].unique():
    for segment in df["Segment"].unique():
        filtered_df = df[(df["Category"] == category) & (df["Segment"] ==
segment)]
        if not filtered_df.empty:
            fig.add_trace(
                go.Violin(
                    x=filtered_df["Category"],
                    y=filtered_df["Profit"],
                    name=f'{category} ({segment})',
                    legendgroup=category,
                    scalemode='count',
                    side='both',
                    line_color='darkblue',
                    points='all',
                    box_visible=True,
                    customdata=filtered_df[['Sales', 'Quantity']].values,
                    hovertemplate=
                        '<b>Category:</b> ' + category + '<br>' +
                        '<b>Segment:</b> ' + segment + '<br>' + # Segment
                        '<b>Profit:</b> ₹%{y:,.2f}<br>' + # Profit
                        '<b>Sales:</b> %{customdata[0]:,.2f}<br>' + # Sales
                        '<b>Quantity:</b> %{customdata[1]}'
                )
            )
fig.update_layout(

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        title="Profit Distribution by Category and Segment (Violin Plot -
Plotly Go)",
        xaxis_title="Category",
        yaxis_title="Profit",
        violinmode='group'
    )
fig.show()

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category_sub_category_sales = df.groupby(['Category', 'Sub-Category'],
as_index=False).agg(
    Total_Sales=('Sales', 'sum'),
    Total_Profit=('Profit', 'sum'),
    Avg_Quantity=('Quantity', 'mean')
)

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color_map = {category: i for i, category in
enumerate(category_sub_category_sales['Category'].unique())}
category_sub_category_sales['color_code'] =
category_sub_category_sales['Category'].map(color_map)

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fig = go.Figure(
    go.Scatter(
        x=category_sub_category_sales["Total_Sales"],
        y=category_sub_category_sales["Total_Profit"],
        mode='markers',
        marker=dict(
            size=category_sub_category_sales["Avg_Quantity"],
            sizemode='area',
sizeref=2.*max(category_sub_category_sales["Avg_Quantity"])/(60.**2),
            sizemin=4,
            color=category_sub_category_sales["color_code"],
            colorscale='Viridis',
            showscale=True,
            colorbar=dict(
                title='Category',
                tickvals=list(color_map.values()),
                ticktext=list(color_map.keys())
            )
        ),
        text=category_sub_category_sales["Sub-Category"],

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        customdata=category_sub_category_sales[['Category']].values, #
        hovertemplate=
            '<b>Sub-Category:</b> %{text}<br>' + # Sub-Category Name
            '<b>Category:</b> %{customdata[0]}<br>' + # Category
            '<b>Total Sales:</b> ₹%{x:,.2f}<br>' + # Total Sales
            '<b>Total Profit:</b> ₹%{y:,.2f}<br>' + # Total Profit
            '<b>Avg Quantity:</b> %{marker.size:,.2f}' # Avg Quantity
    )
)

fig.update_layout(
    title="Sales vs Profit by Sub-Category (Bubble Plot - Plotly Go)",
    xaxis_title="Total Sales (Log Scale)",
    yaxis_title="Total Profit",
    xaxis_type="log"
)

fig.show()

fig = go.Figure()
for category in df['Category'].unique():
    category_df = df[df['Category'] == category]
    for segment in category_df['Segment'].unique():
        segment_df = category_df[category_df['Segment'] == segment]
        symbol_map = {'Consumer': 'circle', 'Corporate': 'square', 'Home
Office': 'diamond'}
        current_symbol = symbol_map.get(segment, 'circle')

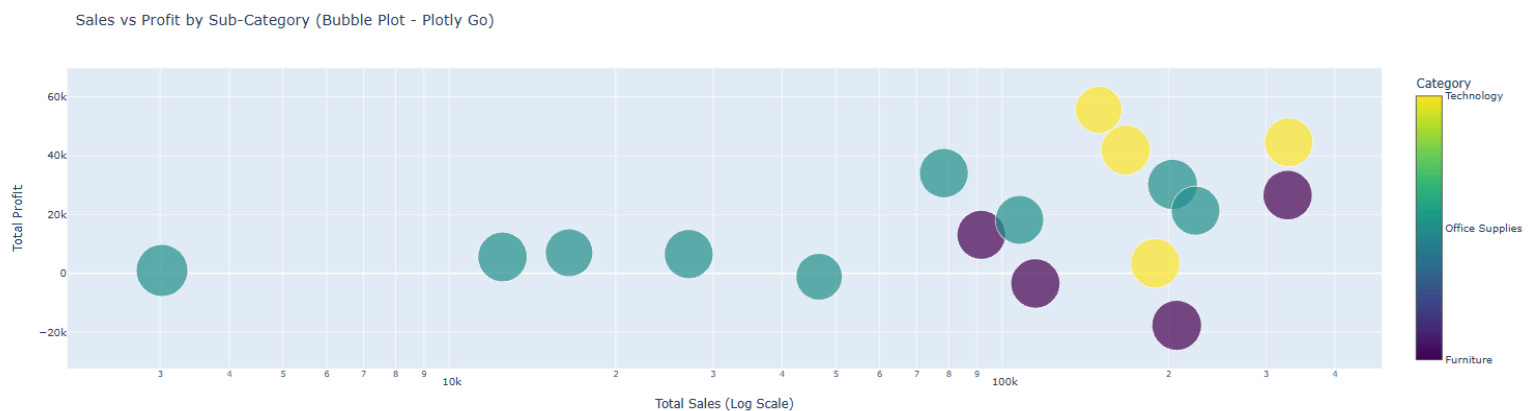
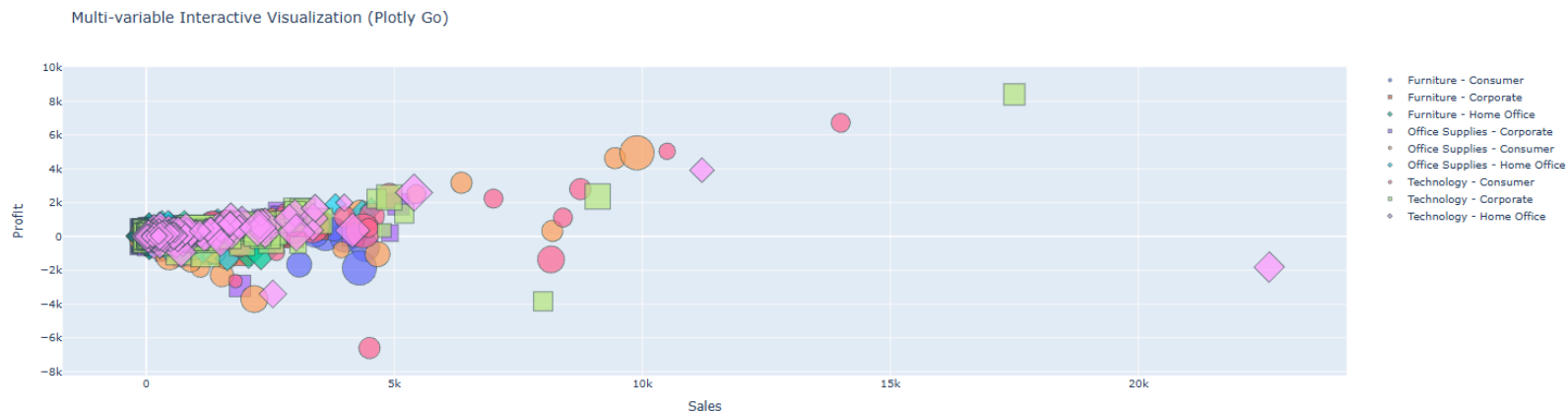
    fig.add_trace(
        go.Scatter(
            x=segment_df["Sales"],
            y=segment_df["Profit"],
            mode='markers',
            name=f'{category} - {segment}',
            marker=dict(
                size=segment_df["Quantity"],
                sizemode='area',
                sizeref=2.*max(df["Quantity"])/(40.**2),
                sizemin=4,

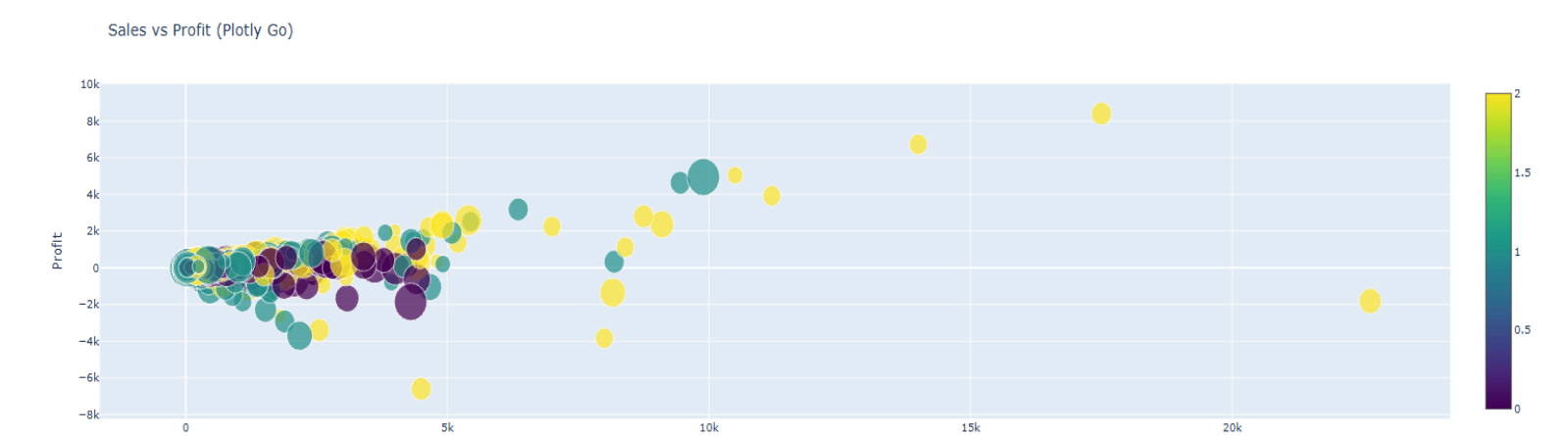
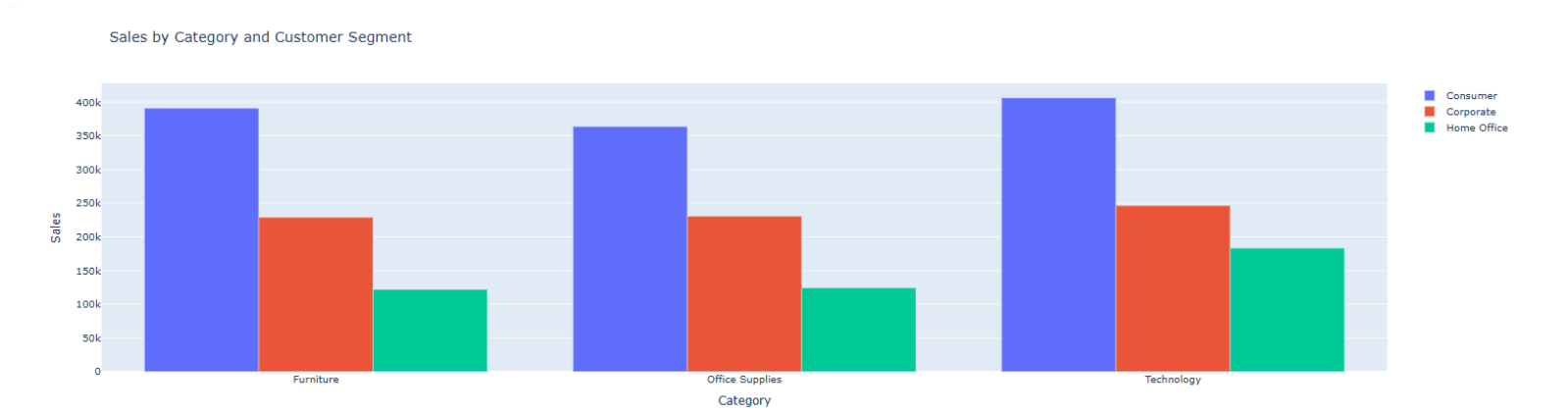
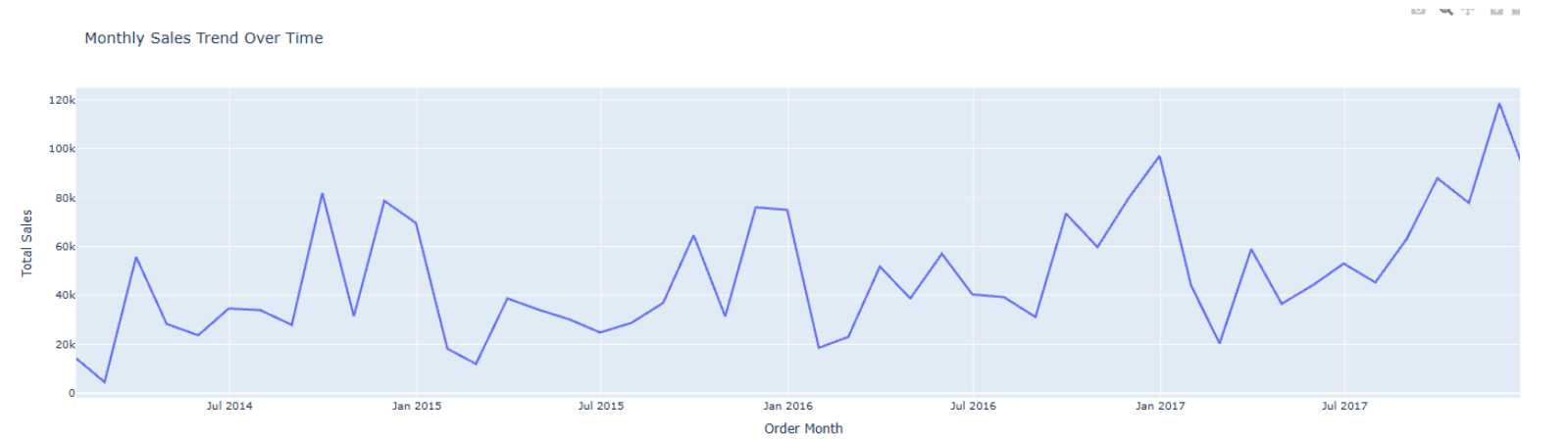
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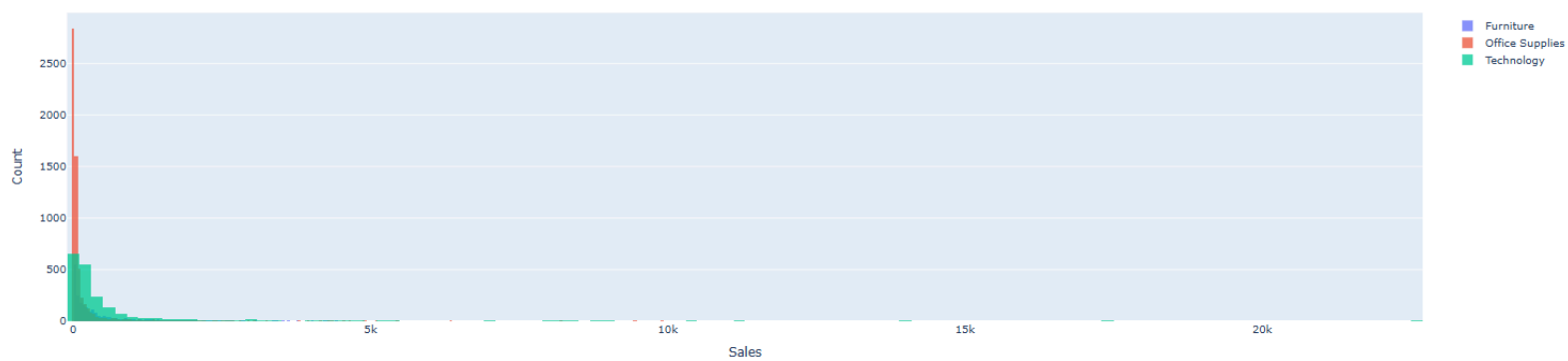
        symbol=current_symbol,
        line=dict(width=1, color='DarkSlateGrey')
    ),
    customdata=segment_df[['Product Name']].values,
    hovertemplate=
        '<b>Product:</b> %{customdata[0]}<br>' + # Product Name
        '<b>Category:</b> ' + category + '<br>' +
        '<b>Segment:</b> ' + segment + '<br>' +
        '<b>Sales:</b> ₹%{x:,.2f}<br>' +
        '<b>Profit:</b> ₹%{y:,.2f}<br>' +
        '<b>Quantity:</b> %{marker.size}' # Quantity
    )
)
fig.update_layout(
    title="Multi-variable Interactive Visualization (Plotly Go)",
    xaxis_title="Sales",
    yaxis_title="Profit",
    showlegend=True
)
fig.show()

```

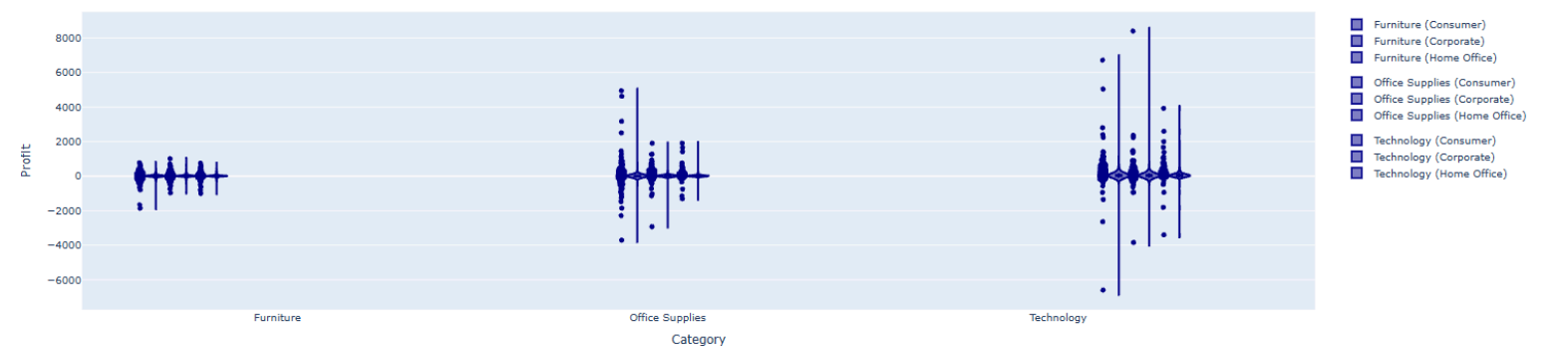




Distribution of Sales by Category (Histogram - Plotly Go)



Profit Distribution by Category and Segment (Violin Plot - Plotly Go)



Profit Distribution Across Categories (Box Plot - Plotly Go)

